**MongoDB Practical: Basic Queries and Operations**

**1.use("student_practical")**

Selects the database for performing MongoDB operations. If the database does not exist, MongoDB creates it when data is first stored.

**Query:**

```
> use("student_practical")
< switched to db student_practical
```

**2. db.createCollection("students")**

Creates a new collection named students in the current database.

**Query:**

```
db.createCollection("students")
```

**Output:**

```
{ ok: 1 }
```

**3. insertOne()**

Inserts a single document into the specified collection.

**Query:**

```
db.students.insertOne({
  rollNo: 101,
  name: "Ram",
  age: 20,
  city: "Nashik",
  branch: "IT",
  marks: 90
})
```

**Output:**

```
{
  acknowledged: true,
  insertedId: ObjectId('6acda55cd6b6110f47c0e911')
}
```

**4. insertMany()**

Inserts multiple documents into a collection in one operation.

**Query:**

```
db.students.insertMany([
  { rollNo: 102, name: "Sita", age: 22, city: "Pune", branch: "IT", marks: 95 },
  { rollNo: 103, name: "Amit", age: 21, city: "Mumbai", branch: "CS", marks: 78 },
  { rollNo: 104, name: "Neha", age: 23, city: "Nashik", branch: "IT", marks: 85 },
  { rollNo: 105, name: "Priya", age: 22, city: "Pune", branch: "CS", marks: 88 }
])
```

**Output:**

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6acda56dd6b6110f47c0e912'),
    '1': ObjectId('6acda56dd6b6110f47c0e913'),
    '2': ObjectId('6acda56dd6b6110f47c0e914'),
    '3': ObjectId('6acda56dd6b6110f47c0e915')
  }
}
```

**5. find()**

Retrieves all documents or documents matching specified conditions.

**Query:**

```
db.students.find()
```

**Output:**

```
{
  _id: ObjectId('6acd18d4aa03cd451840bfa8'),
  rollNo: 101,
  name: 'Ram',
  age: 20,
  city: 'Nashik',
  branch: 'IT',
  marks: 90
}
{
  _id: ObjectId('6acd18e2aa03cd451840bfa9'),
  rollNo: 102,
  name: 'Sita',
  age: 22,
  city: 'Pune',
  branch: 'IT',
  marks: 95
}
{
  _id: ObjectId('6acd18e2aa03cd451840bfaa'),
  rollNo: 103,
  name: 'Amit',
  age: 21,
  city: 'Mumbai',
  branch: 'CS',
  marks: 78
}
```

```
{
  _id: ObjectId('6acd18e2aa03cd451840bfab'),
  rollNo: 104,
  name: 'Neha',
  age: 23,
  city: 'Nashik',
  branch: 'IT',
  marks: 85
}
{
  _id: ObjectId('6acd18e2aa03cd451840bfac'),
  rollNo: 105,
  name: 'Priya',
  age: 22,
  city: 'Pune',
  branch: 'CS',
  marks: 88
}
```

**6. findOne()**

Retrieves the first document matching the specified condition.

**Query:**

```
db.students.findOne({ name: "Ram" })
```

**Output:**

```
{
  _id: ObjectId('6acd18d4aa03cd451840bfa8'),
  rollNo: 101,
  name: 'Ram',
  age: 20,
  city: 'Nashik',
  branch: 'IT',
  marks: 90
}
```

**7. Greater Than ($gt)**

Retrieves documents with field values greater than the specified value.

**Query:**

```
db.students.find({
  marks: { $gt: 90 }
})
```

**Output :**

```
{
  _id: ObjectId('6acd18e2aa03cd451840bfa9'),
  rollNo: 102,
  name: 'Sita',
  age: 22,
  city: 'Pune',
  branch: 'IT',
  marks: 95
}
{
  _id: ObjectId('6acda56dd6b6110f47c0e912'),
  rollNo: 102,
  name: 'Sita',
  age: 22,
  city: 'Pune',
  branch: 'IT',
  marks: 95
}
```

**8. Greater Than or Equal To ($gte)**

Retrieves documents with values greater than or equal to the specified value.

**Query:**

```
db.students.find({
  age: { $gte: 22 }
})
```

**Output :**

```
{
  _id: ObjectId('6acd18e2aa03cd451840bfa9'),
  rollNo: 102,
  name: 'Sita',
  age: 22,
  city: 'Pune',
  branch: 'IT',
  marks: 95
}
{
  _id: ObjectId('6acd18e2aa03cd451840bfab'),
  rollNo: 104,
  name: 'Neha',
  age: 23,
  city: 'Nashik',
  branch: 'IT',
  marks: 85
}
{
  _id: ObjectId('6acd18e2aa03cd451840bfac'),
  rollNo: 105,
  name: 'Priya',
  age: 22,
  city: 'Pune',
  branch: 'CS',
  marks: 88
}
```

**9. Less Than ($lt)**

Retrieves documents with field values less than the specified value.

**Query:**

```
db.students.find({
  marks: { $lt: 80 }
})
```

**Output :**

```
{
  _id: ObjectId('6acd18e2aa03cd451840bfaa'),
  rollNo: 103,
  name: 'Amit',
  age: 21,
  city: 'Mumbai',
  branch: 'CS',
  marks: 78
}
{
  _id: ObjectId('6acda56dd6b6110f47c0e913'),
  rollNo: 103,
  name: 'Amit',
  age: 21,
  city: 'Mumbai',
  branch: 'CS',
  marks: 78
}
```


**12. limit()**

Restricts the number of documents returned by a query.

**Query:**

```
db.students.find().limit(3)
```

**Output:**

```
{
  _id: ObjectId('6acd18d4aa03cd451840bfa8'),
  rollNo: 101,
  name: 'Ram',
  age: 20,
  city: 'Nashik',
  branch: 'IT',
  marks: 90
}
{
  _id: ObjectId('6acd18e2aa03cd451840bfa9'),
  rollNo: 102,
  name: 'Sita',
  age: 22,
  city: 'Pune',
  branch: 'IT',
  marks: 95
}
{
  _id: ObjectId('6acd18e2aa03cd451840bfaa'),
  rollNo: 103,
  name: 'Amit',
  age: 21,
  city: 'Mumbai',
  branch: 'CS',
  marks: 78
}
```


**14. updateOne()**

Updates the first document matching the specified condition.

**Query:**

```
db.students.updateOne(
  { rollNo: 101 },
  { $set: { marks: 96 } }
)
```

**Output:**

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

**15. updateMany()**

Updates all documents matching the specified condition.

**Query:**

```
db.students.updateMany(
  { city: "Pune" },
  { $set: { branch: "IT" } }
)
```

**Output :**

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 2,
  upsertedCount: 0
}
```

**16. deleteOne()**

Deletes the first document matching the specified condition.

**Query:**

```
db.students.deleteOne({
  rollNo: 105
})
```

**Output:**

```
{
  acknowledged: true,
  deletedCount: 1
}
```

**17. deleteMany()**

Deletes all documents matching the specified condition.

**Query:**

```
db.students.deleteMany({
  city: "Nashik"
})
```

**Output:**

```
{
  acknowledged: true,
  deletedCount: 4
}
```